Unlocking the Potential of Deep Learning in Medical Image Analysis

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***Abstract*— Deep learning has transformed illness detection and is assisting the healthcare industry in overcoming obstacles related to correctness and reliability to develop effective as well as resilient computer-aided diagnostic tools. Deep learning approaches provide automated AI-driven utilities that need minimum human oversight to execute tasks associated with the medical detection of fractures, malignancies, internal bleeding, preoperative planning, and intra-operative guiding. Nonetheless, deep learning encounters significant challenges within the burgeoning healthcare sector. This article analyzes the primary obstacles facing the deep learning (DL) research and engineering community in the medical image diagnosis, specifically balanced annotated medical image datasets deficiency, adversarial attacks on deep neural networks (DNN) as well as architectures stemming from noisy medical images, patients' & users' lack of trust, along with ethical and privacy concerns of medical data. This research investigates potential for AI autonomy in healthcare by addressing societal trust issues about autonomous intelligent systems.**

Keywords*— Healthcare, Artificial intelligence, deep learning, diseases, adversarial effect, medical image analysis, convolutional neural network.

I. INTRODUCTION

AI has emerged as a transformative area which has changed remote sensing, healthcare, autonomous driving, robotics, as well as several transdisciplinary sectors in the last decade. Academics and engineers have used machine learning methods to address several biological challenges and basic data processing jobs throughout time [1]. Machine learning (ML) executes a task by acquiring and applying previously gathered, task-specific information [2]. ML algorithms perform well with primary linear information but must be more proficient with complex medical data. With time, DL, a subset of artificial intelligence, has outperformed ML in diagnosing medical imaging because of its enhanced accuracy, capacity to extract deep features, as well as adaptability in addressing complex challenges of diagnosis [3]. Nonetheless, the application of deep learning faces several challenges, including a scarcity of annotated medical image information, degraded medical image data characterized by artefacts as well as attenuations, user skepticism towards DL based models, for example, CAD (computer-aided diagnostic) systems, as well as privacy & ethical issues related to the information [4]. Since these obstacles limit the growth of DL in medicinal field, researchers as well as engineers are significantly troubled by them. Since they provide biased along with erroneous predictions, image artefacts & noise are categorized as adversarial attacks. MRI (Magnetic resonance imaging) scans frequently exhibit motion artefacts, whereas CT (computed tomography) pictures frequently show ring, beam-hardening, scatter, as well as metal artefacts. A combination of Gaussian, Poisson, as well as speckle noise is the most prevalent in radiological medical pictures such as X-rays [5]. Medical profound learning practitioners must manually denoise picture data to provide a clean set of data for testing along with training proposes. More time as well as space is required in this technique, increasing the CAD production cost.

The scarcity of annotated medical picture data necessitates using extensive datasets to test & train resilient DL systems. Labelling of the images is a laborious procedure necessitating oversight by medicinal personnel, resulting in it being costly as well as time-intensive. One major issue with deep neural network training is the lack of data. Furthermore, any imbalance in the dataset's data for a certain class could lead to bias in the DNN, it might result in overfitted model diagnoses as well as predictions. Contemporary DL methodologies utilized representation learning algorithms to produce enormous quantities of medicinal pictures from a limited medical image dataset [6]. Despite effectiveness of artificially created datasets via augmentation in training deep neural networks, skepticism about these data production approaches persists throughout the community [7].

Privacy and ethical considerations significantly hinder the deep learning community's ability to share medical picture data in open-source realm. It is evaluated that the cloud computing value in the healthcare industry will reach 89 billion U.S. dollars by 2027, thus special protocols will be required to safeguard user privacy and stop counterfeiting unless the technology is open source. Many international laws from many nations protect medical data [8]. National regulations were constituted through the HIPAA (Health Insurance Portability and Accountability Act) of 1996 to prevent the disclosure of patients' personal health information without their consent or knowledge. As a result of the legally protected personally identifiable information of patients, healthcare providers must protect, limit, and disseminate it [9]. Consequently, safeguarding patient identity confidentiality is a primary obligation of deep learning engineers, which poses significant challenges. Our study's major goal was to identify and explore the obstacles, potential, explainability, as well as trustworthiness of applying DL algorithms to medical image processing along with healthcare. This study examines:

- The adversarial assaults in deep networks result from attenuation.
- Lack of information and data imbalance.

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- Trust concerns and elucidation.
- Privacy and legal considerations with healthcare information.

II. Deep Learning in Healthcare

Healthcare companies & institutions are increasingly adopting approaches by which artificial intelligence can detect ailments and cure patients. In recent years, deep learning has transformed computers' capacity to interpret and process data with unprecedented speed and accuracy [10]. It is a hierarchical approach that employs intricate and profound structures to learn non-linear data with great accuracy along with precision effectively [11]. DL has been explored in disease detection, processing of biological images, and developing surgical systems for preoperative as well as intraoperative assistance, with promising results. A poll [12] done in the U.S. indicates that individuals possess trust and awareness of AI in healthcare. According to the poll, over half of patients (58%) are familiar with and use various patient-facing healthcare technologies that enable communication with physicians and access to personal medical information. It was emphasized that 52 percent of individuals trust artificial intelligence for their medicinal needs. This indicates an urgent need to build high-performance AI to address various challenges.

Advancement of the AI in healthcare industry necessitates comprehension of medical data, identification of processing methodologies, and utilization of resultant computer-aided diagnoses to deliver precise and dependable outcomes. Accurate comprehension of medical data is essential for successfully using healthcare resources and producing reliable results. After the DL model integrated into the CAD is trained as well as refined by DL engineers, the medical image as input has been accepted by the system as well as produces a diagnostic prediction for customers. CAD system is designed to be resistant to adversarial assaults and mitigations. A severely damaged medical picture may provide an inconclusive diagnostic result for the user [13].

The models' core mechanisms and predictions are transparently accessible to facilitate proper interpretation by human analysts, fostering faith in the forecasts and identifying any flaws or limits in the model. Users contribute their data to the cloud storage of the medical facility to advance CAD research. The development team may receive an inaccurate diagnosis from user for analysis. Furthermore, CADs need to be thoroughly field-tested prior to being utilized by the medical industry, and they need to go through ongoing retraining utilizing real-time data to prevent incorrect diagnosis as well as boost user assurance [14]. The evaluation should occur under the supervision of medical professionals whose experience can authenticate these AI-based CADs and facilitate their enhancement.

A. Data Collection

It includes data collected from secure databases and cloud storage systems (e.g., MRI scans, retinal image capture, X-rays, and CT scans). The data may be real-time or retrieved from an existing image database. The data has been collected from Shaip, an online paid web portal for this study [15].

B. Data Preprocessing and Adversarial Training\

Preprocessing cleans the noise and ensures the quality of the raw medical data. Security mechanisms will train the system to resist adversarial attacks and protect the image's integrity.

C. Model Training and Evaluation

Preprocessed data is used to train deep learning models. Professionals also ensure that the process is running accurately and effectively by checking performance metrics.

D. Prediction and Diagnosis

The trained system makes predictions and diagnoses based on the input data, providing actionable insights for medical professionals.

E. Explainable AI (XAI)

The results are then presented through an explainable AI interface that will help users such as doctors and patients understand and trust the system's decisions. Further data can be shared for ongoing improvement and research.

III. Research Methodology

Figure 1 provides a detailed representation of the research methodology flow, outlining the sequential steps and processes involved in the study.

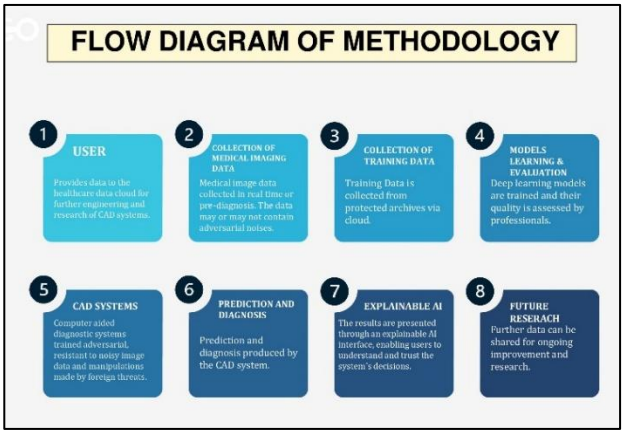


Fig. 1. Diagram illustrating the research methodology steps

IV. Obstacles of Deep Learning in Healthcare

With deep learning, significant transformation may happen in health, yet several obstacles impede its widespread adoption. These obstacles are discussed below:

A. Counterproductive Attacks in Deep Neural Networks Caused by Attenuation

At the time noise is introduced into deep neural networks, they may generate inaccurate outcomes as well as misclassifications [16]. There are several different attenuations along with motion artefacts in medical imaging data; it is not devoid of noise and attenuations. Biomedical imaging techniques such as PET (positron emission tomography), MRI, CT, and ultrasound frequently experience multiplicative noise. There is a noticeable loss of information due to the denoising, which modifies the contrast of the medical image as well as its spatial and temporal distribution. Inadequate protection of AI systems from malicious assaults may engender distrust in AI among general populace. Should any malevolent third-party attempt to execute an adversarial assault, the CADs must be preemptively equipped to address such threats. The inability to withstand these assaults will cause technology users to lose confidence in the CADs. Linear methods are ineffective in eliminating signal-dependent

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multiplicative noise and are characterized mainly by complicated Rayleigh and Gamma models. Moreover, due to high frequencies in the noise and edges of medical pictures, linear filters often exhibit inadequate efficacy in denoising.

DL researchers, engineers, and biologists working within the healthcare decision support system face a huge problem in eliminating artefacts. When machine learning was first applied to the healthcare sector, the image processing community focused on creating pre-processing techniques to eliminate artefacts and attenuations from medical data. When utilized for a smoothed form of the image (context), Liao et al.'s [17] bidirectional filter (CBF) effectively conducts blind image denoising and reduces noise propagation (PoN). Comparing bilateral filtering to linear filtering techniques for example the Wiener filter as well as mean filter, Bhonsle et al. [18] achieved better results on MRI as well as X-ray pictures influenced by additive Gaussian noise with fluctuating variances. A significant amount of research has looked into the utilization of DL for medicinal image denoising. Gondara [19] has demonstrated that convolutional layer block based denoising autoencoders are efficient denoising tools for medical imaging. The author was able to sufficiently train the autoencoder as well as produce denoising results through utilizing a small dataset of only about 300 samples.

Jifara et al. [20] directly learned noise from noisy pictures by combining batch normalization with a residual learning mechanism. Adversarial training has demonstrated increased efficacy in handling lousy data in recent years, even when it does not modify the data to generate outcomes. Yi et al. [21] investigated a diverse array of adversarial training techniques utilizing GANs (generative adversarial networks). Fewer studies on adversarial training have been conducted in the medical industry despite extensive research in the DL field. Adversarial samples, that capture the noisy picture data's general features, improve data, according to the basic idea of adversarial training. The network is built to withstand attacks, including noisy data, produce accurate results, and increase user trust. Exact ground truth is added to the training data [22].

B. Data Scarcity and Unbalanced Distribution

An optimal medical picture dataset must have metadata, an identifier, and a sufficient quantity of photos, precisely annotated with the ground truth by a medical specialist along distributed with the correct license to the DL research community. Metadata consists of annotations as well as data obtained with imaging modalities. The medicinal photographs are, in fact, labor-intensive and costly to acquire. A significant need for extensive, genuine, annotated medical picture data to ensure the DL advancement in medical image diagnostics. Creation of synthetic datasets has started in response to the deficiency of medical picture datasets. This process is called "image data augmentation" including creating pictures from the original dataset via alteration while maintaining the essential characteristics of the image information. The DL algorithm perceives the novel picture as foreign data while simultaneously augmenting the adequate image dataset's volume. A considerable body of investigation has been conducted on data augmentation methods. Fundamental image alteration along with augmentation methods comprises picture translation & flipping, Gaussian blur & noise application, pixel intensity jittering, random cropping, random rotation, affine transformation for shearing, as well as color space transformation. Figure 2 shows the Medical Imagine dataset along with its essential components, challenges, and solutions.

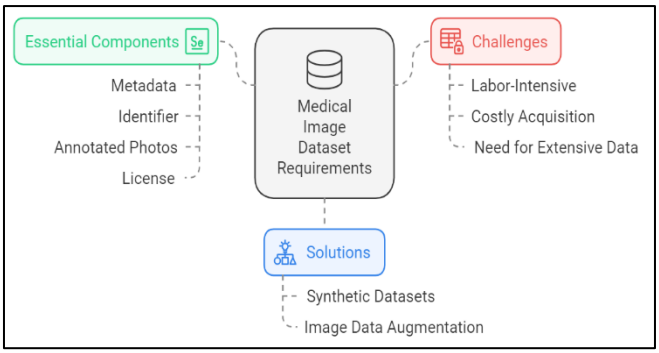


Fig. 2. Requirements of medical imagine dataset

Overfitting occurs when an AI model becomes overly aligned with the training data, which means worse performance on novel data. This phenomenon often occurs when the training data is limited and insufficiently refined, mainly if there is noise involved. Because of the structural similarity to the original data, picture editing techniques do not yield unique images for that feature space, and applying the supplemented data causes the DL model to become overfitted. Some autoencoder and generative adversarial network-based DL techniques are being used to extract the features from the data for every distinct class of an illness, synthesize pictures corresponding to the classes, and help overcome this challenge. The main elements in the Generative Adversarial Network are 2 neural networks: a generator and a discriminator. A GAN generates synthetic data by creating variations from random noise while its discriminator evaluates the generated data about its authenticity. In doing so, training the neural networks against each other will strengthen them in a min-max game. In the GAN-based augmentation, conventionally, the generator learns to generate synthetic medical imaging information from an actual dataset. The generator generates an image dataset for training or validating deep networks for medicinal diagnosis.

Translation-based GANs that employ masks to improve biological as well as medical coherence in visual translation are Pix2pixHD, Pix2pix, & SPADE [23]. Although GANs such as DCGAN [24] exhibit flexibility, PGAN excels at producing high-resolution pictures. Noise-based generation models are frequently employed with data augmentation and patch extractions while managing limited datasets. Sundaram and Hulkund [25] trained and categorized lung diseases, fractures, and pleural effusion types utilizing GAN-generated chest X-ray images. GAN-based augmentation was utilized by Bissoto et al. [26] to improve medical images of skin lesions, and they also examined the diverse synthetic data generated by distinct GANs. Their research demonstrated that StyleGAN 2 [27] surpassed other current models in producing synthetic image data for augmenting skin lesion images. The generation of improved CT images for the classification of liver lesions was carried out by Frid-Adar et al. [28]. It is demonstrated that augmentation can enhance a deep network when a CNN trained with enhanced data from the GAN outperforms a CNN trained on an actual dataset.

Kryuchkov et al. [29] use Conditional GANs on CT images to enhance lung nodules' repeatability and detection likelihood. An open-source CT-GAN network with Wasserstein loss and adaptive instance normalization was employed by the model for data augmentation. Minne et al. [30] carried out tests to compare GAN-based and conventional augmentation methods for enhancing 3D MRI data for

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dementia and Alzheimer's disease diagnosis and therapy. Zhang et al. [31] utilized a domain adaption strategy to segment medical images utilizing meta-learning in order to address the lack of available medical image data. Singh et al. [32] introduced a "MetaMed" methodology to minimize computational demands as well as enhance accurateness with limited sets of data. A 2–5 percent enhancement in the generalization capability had been seen, with 70% accuracy with the hybrid augmentation along with regularization models when assessed on 3 datasets: ISIC 2018, Pap smear, & Break His. Integrating patient trust data and propagation with GANs enhances both patient trust and recommender system efficacy, according to recent promising findings by deep learning researchers on data scarcity and unpredictability.

C. Reliability Concerns and Clarification

Theoretically, DL training algorithms are understandable; the complex mathematics and their structures are similar to a "black box" method. Consequently, forecasting lethal ailments, medical diagnostics for therapy, or else surgical assistance utilizing these AI methodologies engenders ethical apprehensions among individuals. The public tends to distrust these CAD systems along with their assessments of the given data. Consequently, DL sector in healthcare requires extremely skilled professionals to develop those diagnostic tools on the basis of DL. To estimate performance of the model in the classification tasks, common measures include accuracy, precision, recall, as well as the F1 score.

$$Accuracy = (True\ Positive + True\ Negative)/(True\ Positive + False\ Positive + True\ Negative + False\ Negative) \quad (1)$$

$$Specificity = True\ Positive/True\ Positive + False\ Positive \quad (2)$$

$$Sensitivity = True\ Positive/True\ Positive + False\ Negative \quad (3)$$

$$F1\text{-}Score = (2 \times Precision \times Recall)/(Precision + Recall) \quad (4)$$

The model must exhibit robust F1, precision, and recall with commendable accuracy. Sensitivity pertains to the precise identification of those with the condition, while specificity pertains to identifying individuals without the disease. Metrics like accuracy, precision, and others are frequently utilized to evaluate the model's performance on the test set, but they can be modified to represent diverse values. To illustrate the performance of the model AI developers could quickly generate the data. Therefore, customers require greater assurance regarding the model's performance metrics. The absence of trust resulted in the need for XAI that allows customers to comprehend the forecasts as well as diagnoses generated by DL algorithms.

The significant erosion of confidence among society as well as black-box DL models may be rectified using explainable deep learning in the CADs. DL freely processes visual information, in contrast to ML algorithms, and people find visual explainability more compelling and illuminating than numerical metrics. Therefore, for society to embrace AI, CAD systems need to consistently incorporate visual explainability. Utilizing data from the DaTscan (dopamine transporter scan), Magesh et al. [33] examined the interpretability of prediction of the Parkinson's disease utilizing CNNs utilizing the LIME (Local Interpretable Model-agnostic Explanations) technique, and they found encouraging findings. CAMs (class activation maps), guided backpropagation, Grad CAMs (Gradient-based class activation mappings), as well as deconvolution methods have been employed in DL to solve CNN-based picture classification issues. These approaches often belong to the "visual elucidation" domain of DL methodologies. Grad-CAMs, CAMs, and heatmaps were utilized by Cohen et al. [34] to demonstrate how to locate infected areas in a COVID-19 patient's lungs. This allowed for the prediction of the disease's opacity and extent and produced data that could be easily understood. Due to its explainability and illness localization capacities, Gu et al. [35] have successfully tested a total attention convolutional neural network, generating noteworthy outcomes in skin lesion along with the fetal MRI segmentation. Figure 3 displays the various techniques for explainability in medical imaging.

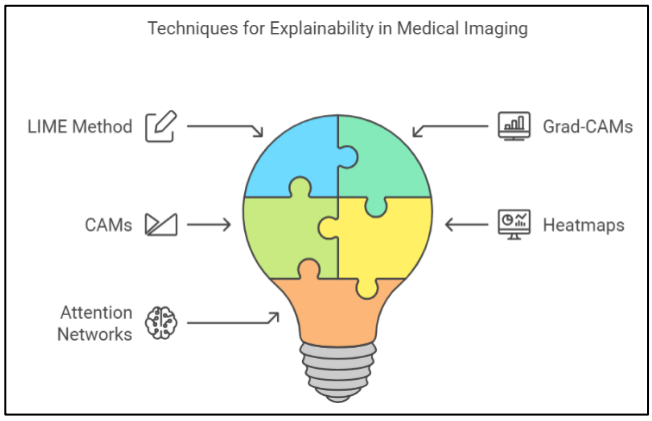


Fig. 3. Techniques for explainability in medical imaging

D. Privacy and Legal Concerns Regarding Healthcare Data

One of the most essential assets in the healthcare system is medical data, which includes all relevant health-related information about patients registered in public or else private medicinal facilities for treatment or else diagnosis. In current years, medicinal information has undergone digitization, and several rules protect its release. The Patient Data Protection Act, referred to as the PDSG (Patientendaten-Schutz Gesetz), had been enacted by the German parliament (Bundestag) in 2020 to thoroughly automate the German healthcare system. Numerous database providers, institutions, and platforms such as U.K. Biobank, Harvard Dataverse, Kaggle, as well as IEEE Dataport, have lately enhanced the accessibility of medical data. They provide licensed data accessible to certified data scientists, ML, & DL algorithms.

Nonetheless, external malicious software may jeopardize these medical databases, necessitating robust cybersecurity measures such as stringent password restrictions, firewalls, and penetration testing, as advocated by antivirus products and cybersecurity professionals. If malicious software can get confidential health information, the patient may experience cyberbullying, harassment, psychological, or else paranoia distress. Therefore, it is up to the data owner or the patient to decide whether to share medical information. Patients often show hesitation because they are concerned about the confidentiality as well as integrity of the data.

A tiny proportion of patients voluntarily provide medical information for research reasons and actively participate in various data collection questionnaires. Accessing sensitive medical data is difficult for DL researchers as well as engineers. Researchers may employ techniques for example differential privacy or pseudonymized data in place of identifiable data to solve these problems. Privacy audits may foster patient confidence, verify acceptable use, and establish security standards to protect against illegal access.

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V. Legitimate Concepts, Possibilities and Future Learning Directions

The realm of DL for medical image recognition encounters the issues mentioned in the previous section, which causes a schism among the DL community and society. AI-driven healthcare systems are not as autonomous as they may be due to societal misinformation and mistrust about intelligent autonomous systems in healthcare and diagnostics. Deep learning systems may be designed with ethical factors like accountability, transparency, explainability, fairness, non-maleficence, and privacy in mind to allay patient worries about data security, confidentiality, as well as integrity [36]. It is imperative to continuously implement measures to sustain the ethical norms of AI across all organizational tiers. The application of ethical AI involves various stakeholders and a number of governance protocols that go beyond technological considerations. Risk assessment, the development of competencies & knowledge, cross-functional cooperation, data governance, stakeholder communication, IT governance, ML Ops, & AI design have been examples of emerging techniques.

A collection of activities utilized to encourage the capabilities, expertise, as well as awareness essential to applying ethical artificial intelligence is called "competence and knowledge development." The collection of communication strategies that firms use to educate stakeholders about their ethical artificial intelligence processes, algorithms, or data is called stakeholder communication. While "data and AI governance" & "AI design and development" both imply a set of operational choices & procedures that businesses utilize to handle moral issues pertaining to the creation, application, and utilization of AI systems, the former term relates to the actual techniques and procedures that go into MLOps. It is an acronym that stands for machine learning operations. Because the development of responsible artificial intelligence systems is of the utmost significance, DL engineers are required to exercise caution while developing deep learning systems that are task-specific. The models need to be thoroughly evaluated for a wide range of data and jobs while protecting the data's security along with eliminating any algorithmic bias that noisy data or foreign assaults may generate [37]. This evaluation must simultaneously be carried out via various performance tests and explainability evaluations. In addition, computer-aided design systems must be equipped with dependable software and be subjected to rigorous penetration testing. Figure 4 illustrates the optimal methods for responsible AI systems to address the shortcomings of traditional AI in healthcare.

Instruments that generate reports on a model's performance and aim to elucidate its processes are referred to as "explainable AI frameworks." Shapley Additive Explanations, commonly referred to as SHAP, is a model-agnostic method that characterizes models through Shapley values originating from game theory. Specifically, it demonstrates how different characteristics affect output or their role in the model's conclusion. Constructing a list of explanations utilizing the Local Interpretable Model-agnostic Explanations tool is possible. These explanations demonstrate how each feature influences the capacity of a data sample for future forecasting. The WIT (Whatif Tool), which Google created, assists users in grasping the functioning of ML trained models, testing one's ability to make assumptions about hypothetical situations, different data properties, subsets of input data, as well as numerous ML fairness metrics [38]. To enhance the interpretability of machine learning algorithms and the explainability of datasets, IBM formulated the AI Explainability 360 toolkit as an open-source initiative

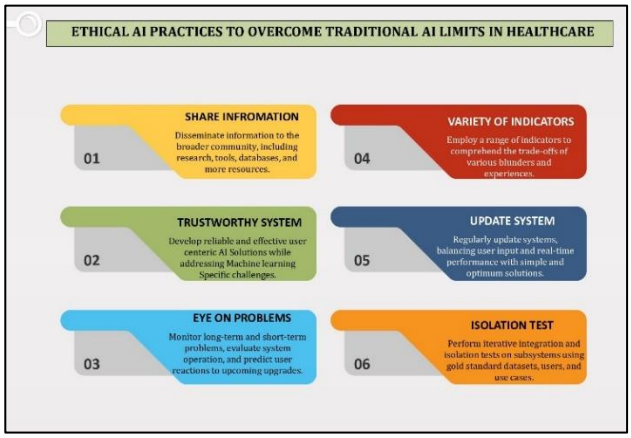


Fig. 4. Best practices for responsible AI

Interpretable ML systems can be more easily developed with the help of the Skater framework. These systems are often put into action in real-world utilization scenarios. Numerous possible applications of XAI exist within the healthcare sector, including but not limited to the following: patient-specific healthcare; disease diagnosis as well as forecasting; medical image segmentation; maintenance and evolution in healthcare technologies; robotic-assisted surgical procedure planning; context-aware systems; basic & translational biomedical research; targeted drug delivery; treatment planning; as well as survival analysis. The research acknowledges the qualitative dimensions of the challenges but neglects the quantitative features of the demographics across the nations and continents where computer-aided diagnosis technologies may be implemented in healthcare. Many rural areas throughout the globe need the means to install such computer-aided diagnosis systems, which is a big obstacle in establishing the dominance of AI in the medical field. Future research may investigate the interactions that take place amongst the many entities that are engaged in AI governance. Governance of information technology, governance of data, and governance of artificial intelligence might be another topic of focus for study in the future. Even though the explainability of artificial intelligence and adversarial training were aspects of medical imaging that had not been investigated before, primary research in these areas has delivered ground-breaking findings. In the world's most remote places with a shortage of medical personnel, computer-aided diagnosis systems integrated with responsible artificial intelligence systems, safe data security rules, and explainable tools might assist healthcare growth. The implementation of new artificial intelligence laws and regulations will be interesting to watch as various companies continue to evaluate the risks along with acceptance of the AI standards, certifications, and audits that are expected to happen in the future.

VI. Conclusion and Future Scope

Though there are obstacles in the way of DL's expansion in medicinal field, the future contains significant opportunities for artificial intelligence in the medical field. These opportunities range from classifying illnesses to investigating AI-supervised robots that can carry out surgical duties. Only

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by regaining people's confidence in artificial intelligence via awareness would it be possible to realize the expansion of DL in diagnosis of medical picture. DL-based designs must be adaptively changed to account for quality degradations that occur over time, just as important as explaining to the user how the system generates the output, how the model is built, and what the model's impact is. In addition to tackling explainability-related concerns, design transparency for DL must be maintained while maintaining user privacy and security and avoiding bias in data collecting, labelling, processing, and model operations. The technological, ethical, as well as sociological dimensions of medical image analysis on the basis of DL may enhance usability and decision making through integrating responsible AI and explainable AI, that comprises the collaboration as well as coexistence of humans & AI. This makes it possible to optimize the designs for better decision-making and accessibility.

The current study lays the foundation for further deep learning-based research on medical image analysis. Future studies can address the challenges highlighted in this study by developing robust methods to improve data quality and handle dataset imbalance, thus assuring equity in healthcare solutions. Additionally, research on XAI techniques will help bridge the trust gap between AI systems and end-users, including patients and health professionals. Eventually, integrating adversarial resilience mechanisms and secure architectures will increase the dependability of deep learning models for critical medical applications. Such a development is also expected from interdisciplinary collaborations with the establishment of ethical frameworks and the proposition of privacy-preserving models for the wide-scale use of AI-driven tools in clinical environments. Further evolution of deep learning in medical image analysis could redefine diagnostic accuracy and improve patient outcomes, besides fostering global healthcare equity.

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